

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own multi-step agent workflows end-to-end using LangGraph, CrewAI, AutoGen, and OpenAI Agents SDK, with tool-calling integrations and stateful agent memory via Redis checkpointing at 99%+ uptime.
- Design ML infrastructure spanning model deployment, evaluation, and data processing: combined GraphRAG with vector-based RAG for grounded retrieval on complex enterprise queries.
- Write infrastructure/systems-level code in Go, Rust, and C++ for GPU cluster scheduling, resource allocation, and fault-tolerant placement logic under hardware-scarcity constraints across AWS, GCP, and CoreWeave.
- Built a deployment-gated evaluation framework enforcing guardrails, model quality checks, and observability across agent/model outputs, blocking 15% of failed releases before production.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Developed multi-modal computer vision models (PyTorch) for medical imaging analysis, and ran hands-on model training/fine-tuning on multi-GPU clusters with compression and quantization for efficient inference.
- Built custom evaluators for RAG/agent workflows tracking drift, latency, and regression across model versions, catching failures before they reached production.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.

Projects

Distributed KV Store & VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a distributed key-value store in C++/Go implementing Raft consensus, linearizable reads, and log compaction; simulated VPC control-plane semantics for network partition and leader-election testing.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++, Go, Rust

Agentic AI & ML Infra: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, PyTorch, Model Training/Fine-Tuning (Multi-GPU), Model Compression/Quantization

Retrieval, Systems & Evaluation: RAG, GraphRAG, Distributed Systems, GPU Clusters (CUDA), Model/Agent Evaluation, Guardrails, Observability

Cloud & Platform: AWS, GCP, CoreWeave, Docker, Kubernetes, PostgreSQL, Redis, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer